The Game:
For this assignment I created a sandbox type game for the Wanderer gamer archetype. There is no “winning” or “points” , just a small world you can play within and explore. The game includes a simple start screen which explains the controls, clicking to drag the “bee” a yellow circle, and keys to place various types of obstacles.

The Bee:
The bee can be dragged around the screen, and thrown to bounce around off the walls. The y-axis of the bee controls the pitch of an oscillator, and the x-axis controls the tone. While held the bee plays constantly, and while thrown the bee makes sound on each bounce. The bee can be thrown at different speeds and angles by dragging faster and in different directions, creating variety in player options for sounds and interactions.

Obstacles:
Further interactions are created by adding obstacles. The most basic obstacle (placed by the space bar) causes the bee to bounce like a wall with no additional sounds. These allow the player to build their own walls, creating differently shaped chambers for the bee to bounce in. Using keys 1-5, the player can place drum obstacles, which play kicks, snares, hi-hats, high toms, and low toms, when the bee bounces off of them. The player uses these to explore different patterns based on where they place them, and how they throw the bee to interact with them.

Break It:
There are ways to “break” the game I have found that result in some of my favorite interactions and sounds from the game. Throwing the bee very quickly results in its location aliasing, where it only exists in its location on the frame, allowing it to jump over obstacles if it does not intersect with it during the frame. This results in some interesting interactions where it will phase in and out of corridors it is bottled into, and move around the map more freely, resulting in more sporadic and exciting movement, and unexpected patterns. If the bee becomes trapped against a drum obstacle, it will hold the envelope open for the filtered noise waveform, resulting in a humming or buzzing sound depending on the filter related to the drum sound. Combining these concepts, the user can throw the bee quickly so it is moving fast enough to alias and then build corridors of different drum obstacles for the bee to become trapped in. This results in the bee phasing in and out of being trapped, resulting in various drum sounds and humming, along with the bee’s own noise, creating an interesting pattern/texture. Other ways of “breaking” the game can be explored by the player, turning this simple sandbox into a versatile playground.